



Digital Twins and Production Lines - A Case Study

BACHELOR'S THESIS

submitted in partial fulfillment of the requirements for the degree of

Bachelor of Science

in

Business Informatics

by

Elias Johannes Hohenwarter

Registration Number 12120485

to the Faculty of Informatics

at the TU Wien

Advisor: Univ.Prof. PhD Henderik Proper

Vienna, May 7, 2025

Elias Johannes Hohenwarter

Henderik Proper

Erklärung zur Verfassung der Arbeit

Elias Johannes Hohenwarter

Hiermit erkläre ich, dass ich diese Arbeit selbständig verfasst habe, dass ich die verwendeten Quellen und Hilfsmittel vollständig angegeben habe und dass ich die Stellen der Arbeit – einschließlich Tabellen, Karten und Abbildungen –, die anderen Werken oder dem Internet im Wortlaut oder dem Sinn nach entnommen sind, auf jeden Fall unter Angabe der Quelle als Entlehnung kenntlich gemacht habe.

Ich erkläre weiters, dass ich mich generativer KI-Tools lediglich als Hilfsmittel bedient habe und in der vorliegenden Arbeit mein gestalterischer Einfluss überwiegt. Im Anhang „Übersicht verwendeter Hilfsmittel“ habe ich alle generativen KI-Tools gelistet, die verwendet wurden, und angegeben, wo und wie sie verwendet wurden. Für Textpassagen, die ohne substantielle Änderungen übernommen wurden, haben ich jeweils die von mir formulierten Eingaben (Prompts) und die verwendete IT- Anwendung mit ihrem Produktnamen und Versionsnummer/Datum angegeben.

Wien, 7. Mai 2025



Elias Johannes Hohenwarter

Kurzfassung

In den letzten Jahren zeigen immer mehr Unternehmen ein starkes Interesse an der Optimierung ihrer Produktionslinien und Fabrikprozesse. In diesem Zusammenhang haben sich Digital Twins und Process Mining als zentrale Technologien erwiesen, um genau dieses Ziel zu erreichen. Digital Twins bieten eine virtuelle Abbildung physischer Anlagen, Systeme oder Prozesse und ermöglichen so eine kontinuierliche Überwachung, Analyse und Simulation in Echtzeit.

Diese Arbeit liefert einen umfassenden theoretischen Überblick über verschiedene Arten digitaler Zwillinge und beleuchtet deren jeweilige Stärken, Vorteile und mögliche Einsatzgebiete in industriellen Umgebungen. Ein besonderer Fokus liegt auf den Datenanforderungen, die für den Aufbau funktionsfähiger digitaler Zwillinge notwendig sind, einschließlich der Methoden zur Erhebung, Integration und Verarbeitung dieser Daten. Zudem wird der Einsatz moderner Simulationswerkzeuge behandelt und wie diese genutzt werden können, um digitale Zwillinge erfolgreich zu entwickeln und zu betreiben.

Durch die Auseinandersetzung mit diesen grundlegenden Aspekten soll diese Arbeit den Leser dazu befähigen, digitale Zwillinge zu verstehen, zu konzipieren und zielgerichtet zur Prozess- und Produktionsoptimierung einzusetzen.

Abstract

In recent years, an increasing number of companies have shown great interest in optimizing their production lines and factory operations. In this context, digital twins and process mining have emerged as essential technologies to support and drive this optimization. Digital twins offer a virtual representation of physical assets, systems, or processes, allowing real-time monitoring, analysis, and simulation.

This thesis provides a comprehensive theoretical overview of various types of digital twins, highlighting their respective strengths, advantages, and potential applications in industrial environments. Special attention is given to the data requirements necessary for building functional digital twins, including methods to collect, integrate, and process this data. In addition, the work discusses the role of modern simulation tools and how they can be used to develop and operate digital twins effectively.

By exploring these foundational aspects, the thesis aims to equip readers with the knowledge needed to understand, design, and implement digital twin solutions tailored to specific production and process optimization goals.

Contents

Kurzfassung	v
Abstract	vii
Contents	ix
1 Introduction	1
2 Enterprise Mining and Digital Twins – A theoretical evaluation	5
2.1 Digital Twin Prototype	6
2.2 Applications of Digital Twin Prototypes in Production	7
2.3 Data Gathering	11
3 Introduction to AnyLogic	15
3.1 Impact on Academic Research	16
3.2 AnyLogic's structure	16
4 Case Study with AnyLogic	23
4.1 Overview of the Case Study	23
4.2 The production layout	23
4.3 Setup in AnyLogic	24
4.4 Running the simulation	26
4.5 Further improvements of the simulation	27
5 Conclusion	29
Overview of Generative AI Tools Used	31
List of Figures	33
Bibliography	35

Introduction

Modern manufacturing companies are increasingly having difficulties designing their manufacturing floors in an efficient way [1]. Due to increasing costs in production, especially in Europe, these companies are interested in maximizing their output with existing resources, such as buildings or staff, which is also discussed in [1]. As we will discuss in more detail below, this is where process mining and digital twins become of importance [2].

Problem Area In the realm of industrial operations, two pivotal concepts are gaining traction: process mining and digital twins [2]. Companies are increasingly interested in these technologies because of their potential to improve efficiency and streamline production. Process mining involves analyzing business processes based on event logs, while digital twins create virtual models of originally physical systems and nowadays also socio-technical systems like production systems.

The main focus of this thesis is on understanding how these technologies can optimize production and improve decision making. A crucial element in this process is the quality of the data, as poor data can lead to wrong conclusions [3]. Because of that, the process of collecting data becomes more important as companies want to use digital twins.

Furthermore, in many industries, production processes are poorly documented, and decisions are often passed on instinct and personal experiences rather than well-founded data. This makes planning difficult because, especially capacity planning, needs accurate data in order to fulfill its task to provide the most efficient utilization of the factory.

Additional hurdles include dealing with data noise, managing outliers, and ensuring that data collection methods are robust and reliable. The need for accurate simulation of production processes without external interference is paramount in addressing these inefficiencies. Such simulations can provide deeper insights into production dynamics, helping to identify bottlenecks and optimize overall capacity planning more effectively.

Ensuring high data quality and implementing precise estimation methods are crucial steps towards enhancing productivity and reducing costs in the manufacturing sector.

Problem Definition The central issue is the growing complexity, because more and more systems play an important role in optimizing production processes, with a significant focus on capacity optimization and bottlenecks. These challenges are increasingly prominent, making it essential to identify and address bottlenecks to improve overall capacity optimization and planning. The manufacturing sector faces persistent inefficiencies that lead to higher operational costs and decreased productivity. Worst case, this can result in investments in new production buildings, although the currently existing one would be suitable for higher outputs, if the available space would be used in a more efficient way, for example. A digital twin of the current production process can help identify inefficiencies, and as a result well-founded decisions can be made on how to use the manufacturing space and resources available at the moment.

Problem Statement My research focuses on creating a digital twin of one production which takes place in a cleanroom. The digital twin will be built with AnyLogic, which is a well-known simulation tool in many branches and industries. The aim is to create a detailed model of the production line, identify bottlenecks, and propose improvements. The research will not consider external factors such as the temperature of the clean room or air quality. The company where the case study will take place is called Lam Research, which is an international company that produces equipment for the semiconductor industry.

Research Questions The overarching research questions is “If and how digital twins and process mining can enhance the production process in manufacturing companies”. In order to provide a structured research approach, the main research question is split into two categories – knowledge questions and design questions.

1. Knowledge Questions:

- What are the benefits of process mining and digital twins in the manufacturing industry?
- How does data quality impact the accuracy of production simulations?

2. Design Questions:

- How can we design a simulation model in AnyLogic to improve the production flow?

Research Objectives & Deliverables The outcome of this thesis should be a theoretical evaluation of digital twins and a simplified production line simulated in AnyLogic with suggestions of what future development could look like in a real-world example. To achieve this goal, two objectives with suitable deliverables are defined.

-
1. **Objective:** What differences are there in regards of digital twins and what are their use cases?

Deliverable: A theoretical evaluation of all different kinds of digital twins.

2. **Objective:** Case Study – Develop a simplified production line with AnyLogic

Deliverable: A simulation model in AnyLogic, capable of running and visualizing the current utilization of the system.

Research Approach/Methodology My research will utilize a combination of process mining and digital twin technologies to gather and analyze data from one production line which was evaluated at Lam Research Villach and then simplified in order to keep confidentiality. The primary tool for simulation will be AnyLogic, chosen for its ability to create detailed models of complex systems and its notoriety in the industry. The research will involve the following:

- Data collection from existing production processes.
- Development of a simulation model in AnyLogic.
- Evaluating current utilization with given input variables.

By following this structured approach, my research aims to provide insights on how to enhance the production process in modern manufacturing companies. Furthermore, the importance of gathering valuable insights and well-founded data about the production process should be emphasized with a theoretical evaluation.

Enterprise Mining and Digital Twins – A theoretical evaluation

According to [4], a **Digital Twin** is a virtual representation of a physical object, entity or product, which is updated with real-time data. The concept has grown more into popularity in the last few years due to the advancements in new technologies like the Internet of Things (IoT), big data and Industry 4.0. The core concept is based on improving the performance and management of physical assets by using computational models of their digital counterparts [4].

As discussed in [4], key components of digital twins include:

1. **Physical Entity (Twin)**: The real-world object or system.
2. **Virtual Entity (Twin)**: The digital replica of the physical object.
3. **Data Connection**: The continuous flow of data between the physical and virtual twins, allowing for synchronization in real-time.

We can distinguish between three types of digital twins [4]:

1. **Digital Twin Instance (DTI)**: This is a digital twin that represents its physical counterpart throughout its entire life cycle. This means, an ever ongoing evaluation of the physical object is taken and every change has a direct impact on the digital twin. It is used to validate the expected behavior and performance of a product or object.
2. **Digital Twin Prototype (DTP)**: When it comes to the manufacturing and production processes of products, a digital twin prototype gathers and stores

valuable information and characteristics about the physical twin. Data can include all kinds of data from computer aided designs (CADs) to information or links to the production chain stakeholders. This type of DT is able to simulate manufacturing scenarios and perform validation testing, evaluation, and even quality control testing prior to the actual manufacturing process itself.

3. **Performance Digital Twin (PDT):** This type of DT is able to monitor, aggregate and analyze data of products in a more real and unpredictable environment. The PDT can provide design optimizations, maintenance strategy generation and drawing conclusions from a product's performance [5].

Since the case study of this thesis is about production simulation and finding ways to improve the performance of a production line what includes increasing output and removing bottlenecks, the Digital Twin Prototype will be used.

2.1 Digital Twin Prototype

A Digital Twin Prototype can be achieved through six essential steps and components [6]:

1. **Virtual Representation of the physical system:** A Digital Twin Prototype is essentially a virtual model of a physical system. This virtual model can include emulators for sensors and actuators. These emulators include predefined configurations in order to simulate the real-world behavior.
2. **Emulators for physical components:** Software emulators are used to replace and simulate physical hardware components such as sensors. This allows the software to interact with the virtual twin as it would be the actual hardware, enabling changes in the system without the need to change the physical system.
3. **Scenario testing without the physical connection:** The Digital Twin Prototype operates completely independent of the actual physical system. This allows engineers to run various scenarios and observe the behavior of the system.
4. **Incremental development and continuous integration:** By integrating the outcomes of the Digital Twin Prototype into the development or continuous improvement workflow, autonomous testing can be implemented. In other words, changes to the code base can be quickly tested.
5. **Modular and scalable design:** These types of DTs are designed to be modular and scalable. This means that as the system changes, the DTP can be quickly adopted to reflect additional components or improved features.
6. **Integration with software development pipelines:** Integrating the Digital Twin Prototype in automated testing chains with containerization technologies create consistent testing environments throughout multiple stages of the development cycle.

2.2 Applications of Digital Twin Prototypes in Production

Digital Twin Prototypes (DTPs) provide the ability of virtual representations of a physical system, products and processes. By integrating real – world data, DTPs empowers engineers and production managers to simulate, monitor and optimize various aspects of a production line. The following section dives deeper into the diverse applications of Digital Twin Prototypes in production environments.

2.2.1 Virtual Prototyping and Product Design

Virtual prototyping and product design are two of the most impactful uses of DTPs. Creating a digital representation of a product before physical manufacturing enables engineers and product designers to test and validate their concepts.

Virtual prototyping allows for:

- **Design Iteration:** As mentioned in [7], DTPs allow rapid testing of multiple iterations of a product or design to achieve optimal performance.
- **Material Analysis:** Identifying the best material for production, including behavior under stress and different environmental conditions is also one capability of DTPs according to [8]
- **Cost Reduction:** Another major points of DTPs is cost saving. The paper [9] mentioned the reduction in costs because no physical prototypes are needed to be built, which is often time consuming.

In short, Digital Twin Prototypes are capable of simulating real-world scenarios such as stress testing, aerodynamics, thermal performance and many more, ensuring that the final design meets the desired specifications. As discussed in [7], this leads to fewer errors during the actual production process and faster time-to-market for new products.

2.2.2 Production Process Optimization

Digital Twin Prototypes are at their best in modeling and optimizing entire production processes. Through simulation, manufacturers can analyze and improve workflows, identify inefficiencies and therefore minimize waste and energy consumption.

- **Bottleneck Identification:** As mentioned in [7] detecting production inefficiencies is a key strength of DTPs.
- **Energy Optimization:** Saving energy usage costs is also possible with DTPs [7].
- **Quality Improvement:** [10] also mentioned a consistent product quality due to the usage of a Digital Twin Prototype.

Summarized, a DTP can simulate the impact of changing machine parameters, material properties or production speeds on output. Running these simulations leads to the identification of the best strategy for certain scenarios.

2.2.3 Predictive Maintenance

DTPs also provide significant advantage in facilitating predictive maintenance. Potential issues can be identified before they lead to equipment failure. This is possible by collecting real-time data from production machines and feeding it into the digital twin.

- **Real-Time Monitoring:** All production equipment is continuously observed through sensors observing their performance [11].
- **Failure Prediction:** The paper [12] mentions the identification of patterns and trends in order to detect signals for potential breakdown of the production line.
- **Cost Reduction:** These points, again, are leading to cost reduction.

For example, a DTP might simulate the wear and tear of a conveyor belt, allowing maintenance teams to replace it proactively before it disrupts production.

2.2.4 Workforce Training and Simulation

Workforce training and simulation is a topic which seems odd at first sight, but it provides a safe place for the first experiences and touching points with a production processes without being exposed to dangerous situations or hazardous substances. Applications in workforce training include:

- **Immersive Learning:** Operator can interact with a digital representation of equipment in a virtual space. [7]
- **Scenario Testing:** [12] also mentions the simulation of emergency situations to train staff on appropriate responses as a key feature of DTPs.
- **Process Familiarization:** Another key feature of DTPs according to [7] is helping new employees understand workflows and production processes without disrupting the actual production.

This means that providing a realistic simulation of the production environment can reduce learning time and improve the overall readiness of the workforce.

2.2.5 Supply Chain Management

In addition, Digital Twin Prototypes play a crucial role in optimizing supply chain management. By modeling the flow of materials and products throughout the supply chain, manufactures can achieve greater efficiency and responsiveness. This also applies to the internal flow of material. For example, material from warehouse to production. Key applications include:

- **Demand Forecasting:** Including real-time and historical data to predict demand and adjust production schedules accordingly [7].
- **Inventory Management:** By simulating inventory levels, overstocking and stock outs can be prohibited.
- **Logistics Optimization:** Planning transportation routes and schedules can minimize delivery times and costs.

Therefore, a DTP can simulate the impact of a delayed shipment of raw materials on the entire production line. This makes it possible to adjust production plans in advance to avoid disruptions.

2.2.6 Sustainability and Environmental Impact

As sustainability and environmental impact becomes more and more of importance all over the world, DTPs can help optimizing resource usage and minimizing wast. Simulating the energy consumption of a production line enables the identification of areas where savings can be achieved [13].

Digital Twin Prototypes (DTP) are a transformative tool in modern manufacturing. This allows real-time virtual representation of physical systems. Their application covers several key areas:

- Virtual prototyping
- Increasing production efficiency
- Predictive maintenance
- Employee training
- Supply chain management helping manufacturers model, track and improve various aspects of production precisely by leveraging real-world data.

Important uses:

- Virtual prototyping and product design: Engineers can test designs in multiple iterations.

- Simulate material behavior and reduce the dependency on expensive physical prototypes. This will help speed up the development cycle and improve product quality.
- Manufacturing Process Optimization: DTP models the entire workflow to identify bottlenecks. Increase energy efficiency and maintain quality control. Helps increase efficiency and production.
- Predictive maintenance: Real-time machine monitoring allows for early identification of potential failures. Reduce unplanned downtime and extend the life of the equipment.
- Workforce Training and Simulation: The virtual environment provides a risk-free platform for operators to gain hands-on experience. Simulate emergency response and understand the work process without affecting production.
- Supply chain management: By modeling material flow and logistics, DTP optimizes inventory levels. Transportation schedule and demand forecasting to optimize the supply chain.
- Sustainability: Simulation of energy use and waste reduction processes supports environmentally friendly practices and aligns with global sustainability goals.

This thesis is mainly focused on simulating internal production processes, not considering any other possible area of application of Digital Twin Prototypes discussed earlier. The primary reason for this is controllability. Internal production processes are completely under the manufacturers control, allowing for precise data collection and targeted optimization.

External factors are important, but often include unpredictable variables that are difficult to control or replicate accurately in a simulation. For example:

- disruption in the supply chain,
- market demand shifts or
- external economic conditions

are subject to a brought influence outside of the manufacturing environment. Excluding these factors allows the study to focus on production performance metrics.

Simulation models that only focus on the internal dynamics of production lines provide immediate and measurable benefits building a base for future studies, which are not the scope of this thesis. The scope of this research aligns with the industry needs in which manufactures prioritize improving internal processes to remove bottlenecks and increase production output.

By limiting the focus to production simulation, this research will maximize the potential for tangible improvements within a controlled environment. This will ensure that the benefits of the Digital Twin Prototype are fully realized in addressing critical operational challenges.

2.3 Data Gathering

Data gathering is a critical step in the creation and implementation of a Digital Twin Prototype (DTP). It involves the collection of relevant and accurate data from the physical production environment, which serves as the foundation for building a reliable and functional virtual representation. The quality and scope of the collected data directly influence the precision and effectiveness of the digital twin.

This chapter focuses on the methods and tools used to gather data for simulating a production line. The data required typically includes machine performance metrics, sensor readings, production outputs, and workflow parameters. Collecting this data involves integrating various technologies such as IoT devices, sensors, and real-time monitoring systems, to ensure continuous and accurate data flows.

The objective of this chapter is to outline the processes for identifying, acquiring, and processing the data needed to develop a Digital Twin Prototype. By ensuring robust data gathering practices, the DTP can accurately reflect the physical system, allowing realistic simulations and meaningful insights to improve production efficiency. The subsequent sections will discuss the types, sources, collection methods, and challenges associated with collecting data for production simulations.

2.3.1 Data Types and Sources

The data required for a DTP typically includes:

- **Machine Performance Metrics:** Various data from equipment, such as efficiency, operational status and output rates.
- **Sensor Readings:** Information from IoT devices, which are monitoring variables like temperature, pressure or any other industry specific value.
- **Production Outputs:** Records of product quantities, quality metrics and defect rates.
- **Workflow Parameters:** This can include details on process sequences, timing and resource utilization.

To utilize the data, it is of great importance to have a look at data collection methods.

2.3.2 Data Collection Methods

There are a lot of effective data collection methods available.

- **Sensor Integration:** Deploying and integration sensors to capture real-time data from machinery and processes.
- **Data Acquisition Systems (DAS):** Utilizing DAS to aggregate data from multiple sources for centralized analysis.
- **Cloud-Fog-Edge Computing Collaboration:** Implementing a hierarchical computing approach to manage data collection, storage and processing efficiently [14].

2.3.3 Importance of Data Quality

High data quality is essential for the success of a digital twin. Inaccurate or incomplete data can lead to wrong simulations and therefore not optimal decisions. In [14] are several factors discussed to ensure data quality:

- **Data Validation:** Implementing checks to verify data accuracy and consistency.
- **Data Integration:** Combining data from various sources into a cohesive dataset.
- **Data Management:** Establishing protocols for data storage, access and security.

The reference [14] also discusses common challenges in data gathering, such as:

- **Data Heterogeneity:** Includes managing diverse data formats and sources.
- **Data Volume:** Handling large datasets generated by continuous monitoring.
- **Data Latency:** Ensuring real-time data availability for in-time decision making.

A structured approach to data collection and management ensures that the digital twin accurately reflects the physical system.

2.3.4 Collecting Data “by Hand”

However, in some cases, a detailed description of a production process is not available, either due to a lack of documentation, a lack of digital systems or limitations of the production process. In such scenarios, data collection must be done manually, to gather the necessary data for the DTP. Although this approach is time consuming and labor intensive, it can provide a very granular understanding of the production process.

Observing and Process Mapping

Direct observations include watching and recording the entire production process.

- **Process Flow Mapping:** This technique helps to identify inefficiencies and areas of improvement by documenting the sequence of operations.
- **Timing Studies:** Measuring the duration of specific tasks or cycles to establish baseline performance metrics. This can also include taking times, which can be done using the MTM approach. MTM stands for Method-Time-Measurement and is used to track and improve human work steps, which is explained here [\[15\]](#).
- **Resource Utilization Tracking:** To optimize resource allocation, information about usage of material must be collected.

Interviews with Operators and Engineers

Operators and engineers possess practical knowledge that may not be documented but is crucial to understanding the nuances of the production line. Key aspects include:

- **Structured Interviews:** Interviews with predefined questions ensure that all relevant topics are covered.
- **Informal Conversations:** Engaging in casual discussions and conversations can encourage openness and may reveal relevant insights.
- **Focus Groups:** Group discussions between several departments can highlight systemic issues that affect the production process.

This data collection approach is the main source of information for the case study conducted in this thesis. The production process is largely done by hand and therefore no fancy IoT devices or modern sensors can collect the actual production process in an accurate way.

Introduction to AnyLogic

“AnyLogic is the leading simulation modeling software for business applications, utilized worldwide by over 40% of Fortune 100 companies. AnyLogic simulation models enable analysts, engineers, and managers to gain deeper insights and optimize complex systems and processes across a wide range of industries.” [16]

Simulation modeling has become an essential methodology in both academic research and industry to analyze, understand, and optimize complex systems. AnyLogic [16] describes itself as, among various simulation tools available, outstanding. AnyLogic is a versatile and comprehensive platform due to its ability to integrate multiple modeling paradigms within a single framework. Developed by the AnyLogic Company, it is widely recognized for its flexibility in representing dynamic systems using agent-based, discrete event, and system dynamics modeling. This multi method approach enables researchers and practitioners to simulate and study intricate real-world problems that span multiple domains, including supply chain management, healthcare, manufacturing, transportation, and social systems.

As discussed in [17], the multi method approach of AnyLogic distinguishes it from other simulation tools that typically focus on a single method. Discrete event simulation (DES) allows researchers to model systems as networks of queues and events, suitable for manufacturing and logistics processes where system states change discretely over time. As also mentioned in [17], Agent-based modeling (ABM) is ideal for systems where individual agents (e.g., people, vehicles, organizations) interact and adapt to changes in their environment. System dynamics (SD) enables the modeling of continuous processes and feedback loops, which is particularly valuable for studying long-term behaviors in areas such as climate change, economics, and public health [17].

The ability to combine these methods in a single model provides researchers with the flexibility to choose the most appropriate technique for each part of a system. For example, in healthcare research, DES might be used to simulate patient flow through

a hospital, while ABM could simulate how individual patients respond to a pandemic outbreak, and SD could model the long-term effects of policy changes on healthcare systems. [17]

3.1 Impact on Academic Research

According to [18] AnyLogic has become an indispensable tool for academic research, not only due to its versatility but also due to its ability to model real-world complexities across multiple domains. Whether used for educational purposes or for cutting-edge research, AnyLogic allows scholars to test hypotheses, optimize processes, and predict outcomes in a controlled, risk-free environment, as discussed in [18]. Its capacity to integrate DES, ABM, and SD makes it particularly valuable for interdisciplinary research, where problems often span multiple fields and require varied methodological approaches [18].

The continuous development of the platform, evidenced by its recent updates, ensures that AnyLogic remains at the forefront of simulation technology. The 2023 AnyLogic Conference showcased significant advancements in performance and usability, including new modeling capabilities and enhancements to the cloud-based experiment framework. These developments promise to further extend the software’s utility in research and industry alike. [19]

3.2 AnyLogic’s structure

In order to get a basic understanding of the software AnyLogic, the following will provide an overview of the software components which will be mainly used to fulfill the case study. The AnyLogic software version which was used for the research in this thesis was 8.8.4 Professional.

On the left side of the window, there are two ribbons. The first one is called “Projects” where all the currently open projects are visible. Furthermore a deeper look at the projects properties and components can be achieved here, such as included components and agents.

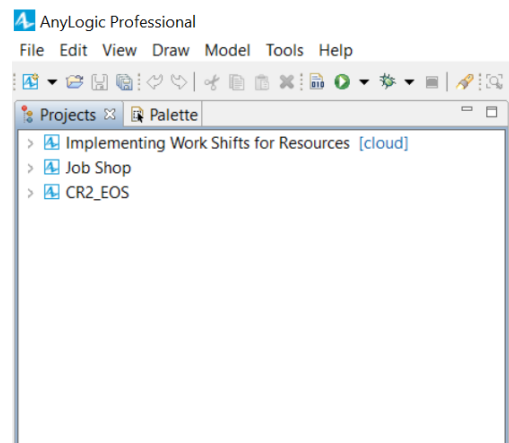


Figure 3.1: Projects ribbon

The second ribbon is called “Palette” where several libraries are available. The purpose of these libraries is to provide process modeling components, visual elements or special

agent properties just to mention a few of them. Of most importance for the case study are the “Process Modeling Library”, “Material Handling Library”, “Agents”, “Space Markup” and “Presentation”.

Process Modeling Library The Process Modeling Library provides elements, which are needed to model a process like in a flowchart.

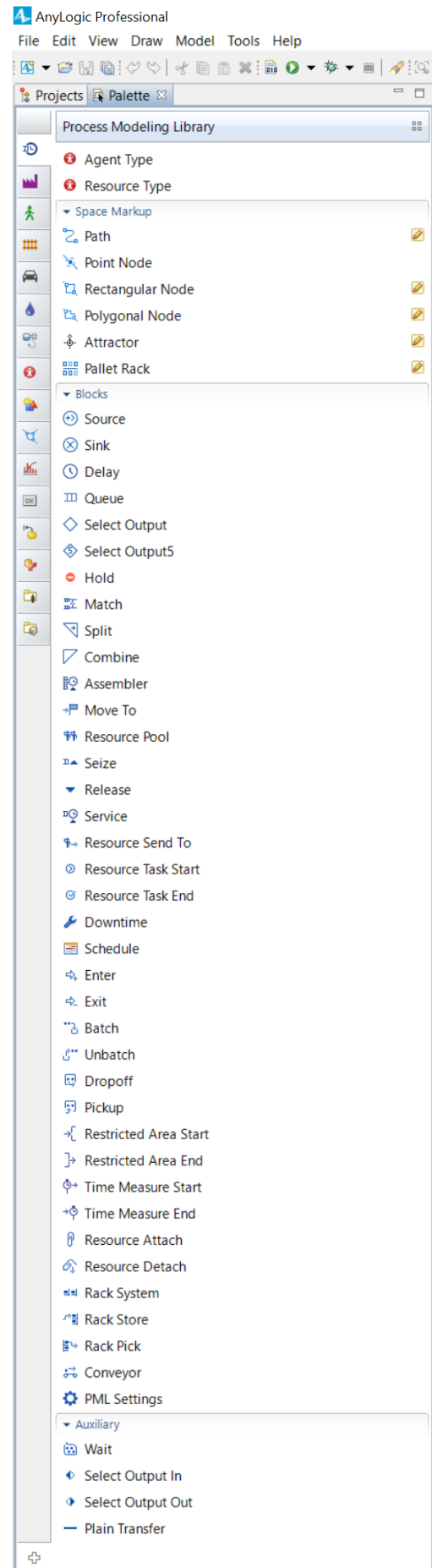


Figure 3.2: Process Modeling Library ribbon

Material Handling Library The Material Handling Library provides elements to simulate and model the flow of material throughout the entire manufacturing process, such as retrieving items from the warehouse or moving material with a transporter.

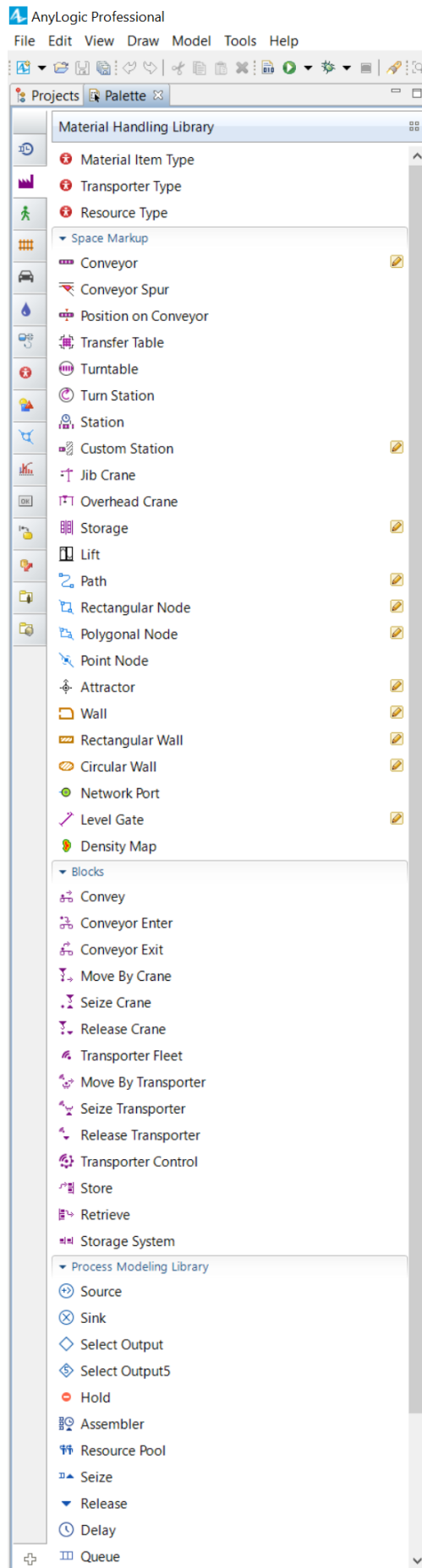


Figure 3.3: Material Handling Library ribbon

Agents The Agents section contains everything related to special elements, which can include visual representations, variables, collections or some other specific detail. These container elements are called Agents in AnyLogic.

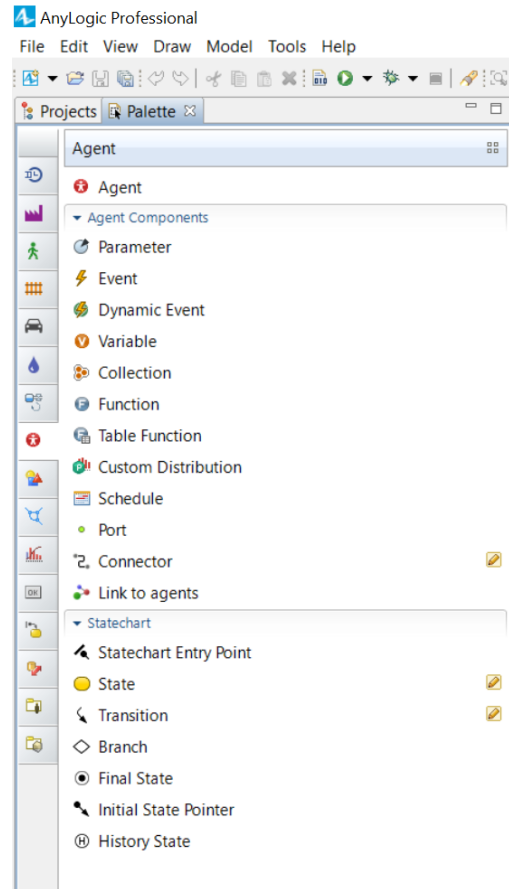


Figure 3.4: Agent ribbon

Space Markup This section provides elements like nodes or paths, as well as more specific items like a conveyor or an overhead crane.

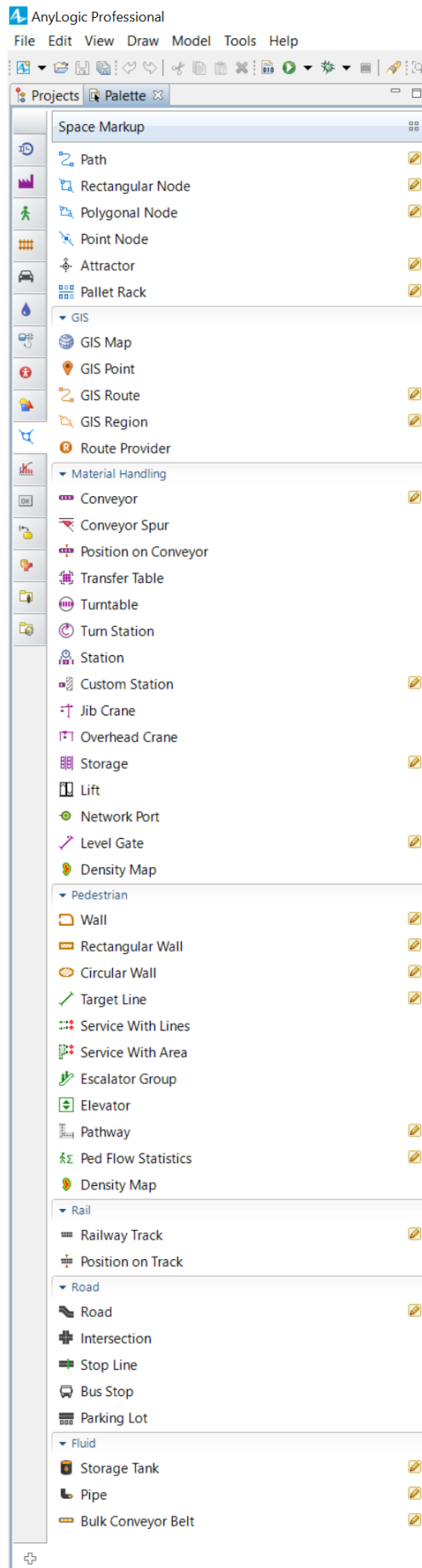


Figure 3.5: Space Markup ribbon

Presentation The last section used for the case study is the Presentation section. It contains everything needed for an appealing visual representation of the system. This includes elements such as images, text or special lighting.

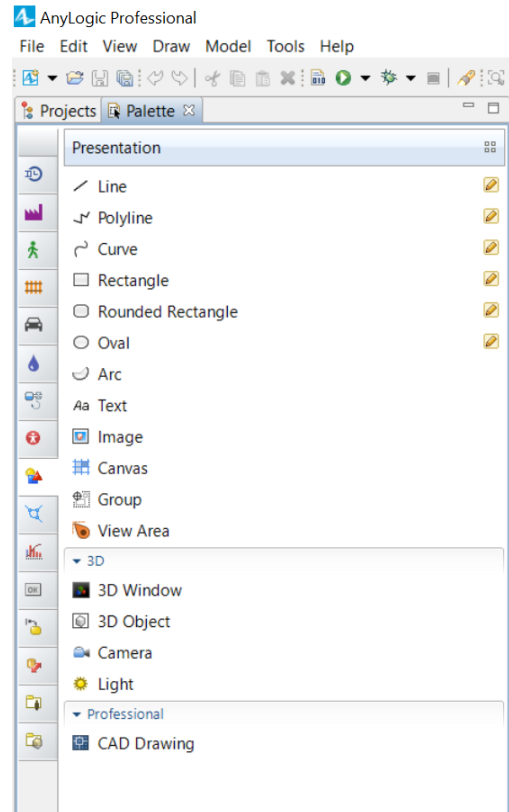


Figure 3.6: Presentation ribbon

Case Study with AnyLogic

In order to underline the theoretical part of this thesis, a case study with AnyLogic was conducted.

4.1 Overview of the Case Study

The scenario which is evaluated with AnyLogic is a production with one production line. This production line starts with three input lines and merges into one output. The key challenge here is that the three input lines need different amount of time to complete and interim storing space is very limited in production. Furthermore, AnyLogic should be able to prove or refute the intuitive first thought of a production planner about how to improve throughput. The goal of the simulation should be to provide an brief look at how a digital twin could be built and what challenges are on the way. The simulation will not be fed any live data. All input and process rates are scheduled within the simulation in order to be able to evaluate different scenarios.

4.2 The production layout

The simulation is based on a simplified production line from Lam Research Villach that is located in a cleanroom.

4. CASE STUDY WITH ANYLOGIC

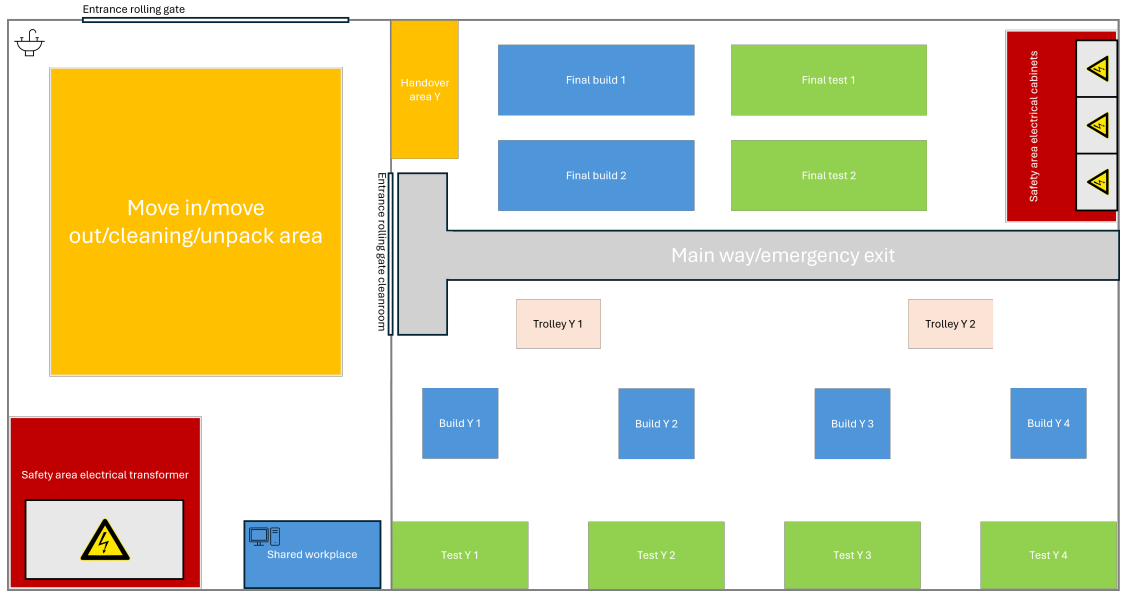


Figure 4.1: Production layout

As visualized above, there is enough space available to bring raw products in parallel into the production, which will be called clean or unpack in the AnyLogic simulation. Furthermore, enough capacity exists to build four raw products of Y in parallel and there are two assembly workplaces to complete two final products in parallel. The same situation exists for the testing area.

4.3 Setup in AnyLogic

First of all, the initial setup was defined. As mentioned earlier, the production line consists of three separate input products, X, Y and Z. These inputs need to be processed at first in order to be combined and assembled.

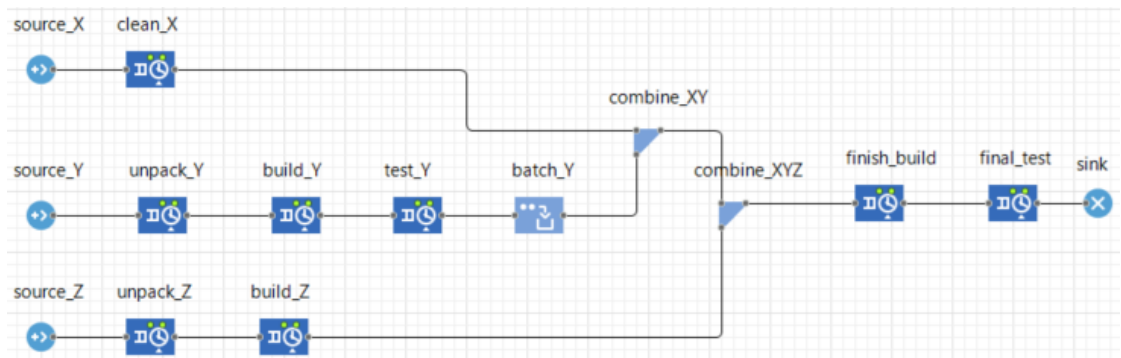


Figure 4.2: Initial setup in AnyLogic

After that, resource pools were inserted for the service blocks. These resource pool blocks tell the service blocks when they are active and for how long.

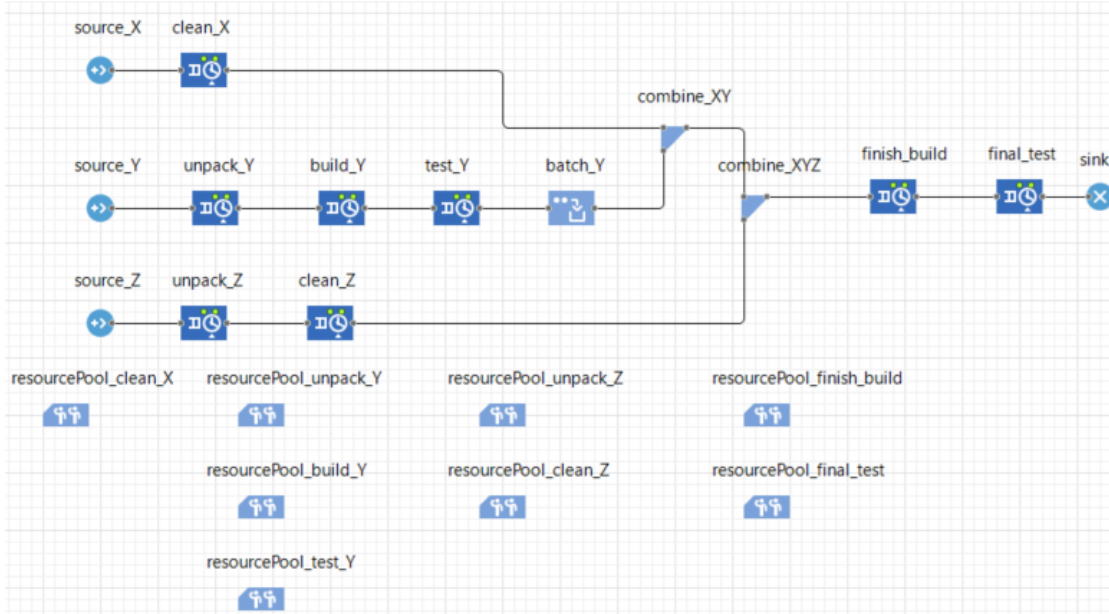


Figure 4.3: Added resource pools

In order to complete the setup, schedule blocks were added in order to tell the resource blocks when they are active and for how long. Furthermore, there are also rate schedules for the source blocks. These arrival schedules are used in order to be able to clearly define when and how many inputs of X, Y and Z should arrive in the model.

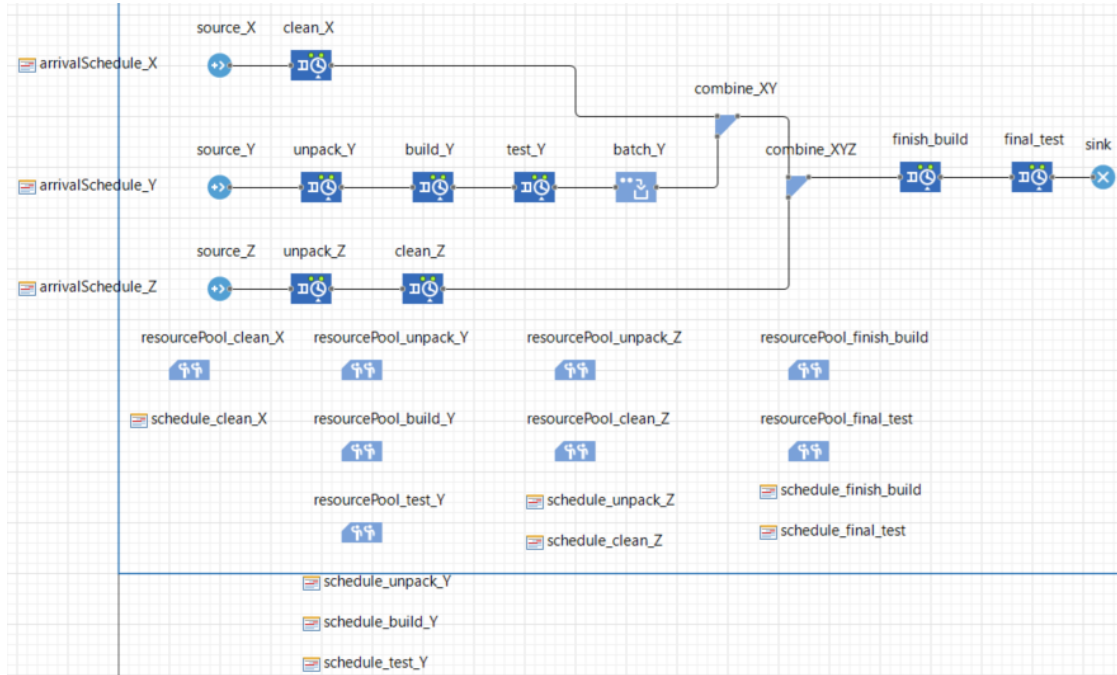


Figure 4.4: Added schedule blocks

4.4 Running the simulation

After completing a simple setup of a production line, it is now possible to run the simulation. Due to the schedule blocks, the ability was created to set different input and rate schedules. Two work shifts per day were defined. The first one from 4 a.m. to 1 p.m., then production takes a short break until the second shift starts which goes from 2 p.m. until 11 p.m.. The input of material into the production line is also scheduled with the help of service blocks. X is brought into the production one time per shift. From Y there is the need for four items in order to complete the final product, therefore, the input in production of Y is four per shift. Z is needed once to complete the final product, so the input in production is one time per shift. The input schedule occurs at the beginning of a shift. This should ensure that there is always “work to do”.

Due to a technical problem, the simulation was stuck with building four inputs of type Y at once. The service block build Y always built just one item at a time, although the resource pool and the schedule were configured to give it four available slots per work shift.

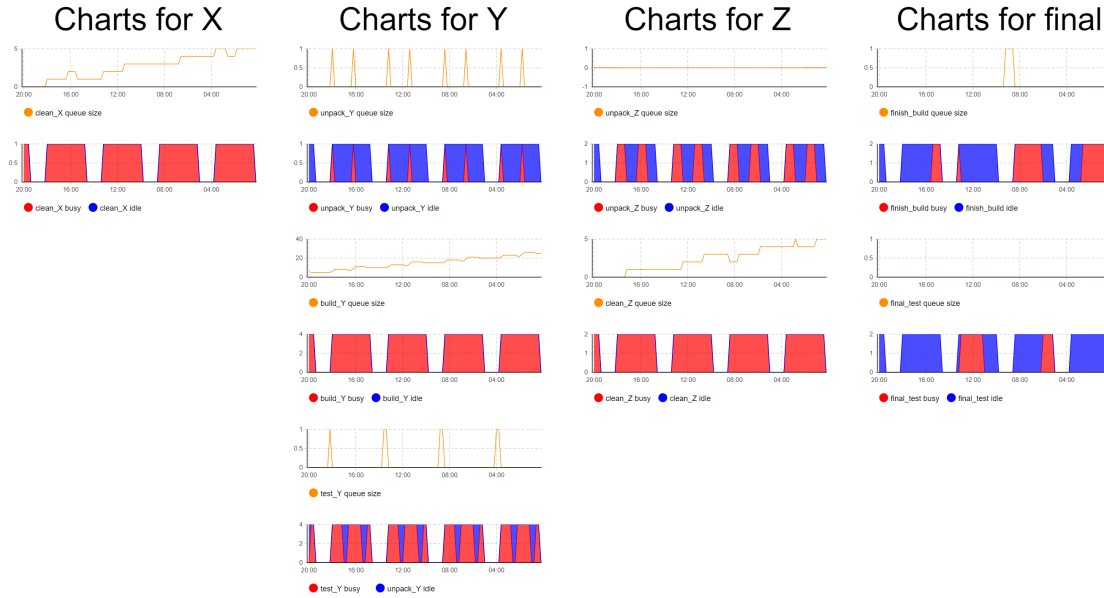


Figure 4.5: Charts

This problem is clearly visible in the chart, representing the queue for the build Y service block. The growth of the queue is also affecting all other blocks, which can also be observed in the figure above.

4.5 Further improvements of the simulation

In order to have a real digital twin real time data would be needed. AnyLogic supports this feature and allows SQL selects. Due to the technical problem above this feature is only discussed theoretical. In order to achieve a simulation with real time data a SQL select can be executed for receiving data from a database. This data can then be used to feed the schedule or resource pool blocks of the simulation. Depending of the use case of the simulation, multiple selects over time can be executed by implementing an event block into the simulation. This allows the simulation to be always up to date and running endlessly over a longer period of time. Of great importance here is, that the data must be in consistent good quality in order to keep the simulation running, unexpected values or outliers can break the simulation and lead to unexpected behavior which may lead to false conclusions. Additional checks and fail-saves should be implemented before data enter the database in order to prevent this from happening.

Conclusion

As demonstrated throughout this thesis, digital twins are set to gain increasing relevance in modern production environments. They offer significant improvements in key performance areas such as throughput, output quality, process reliability, and operational efficiency. From real-time monitoring of individual production lines to the simulation of entire factories, the potential use cases for digital twins are vast and continuously expanding.

Moreover, digital twins can play a crucial role in employee training, allowing workers to interact with virtual models of real systems in a risk-free environment. With the right understanding of production processes and proper training, the implementation of digital twins becomes not only feasible but also highly cost-effective. The initial effort required is relatively low compared to the long-term benefits they can bring, particularly in terms of cost reduction, process optimization, and increased flexibility.

In conclusion, it is evident that digital twins have the power to fundamentally enhance the efficiency and adaptability of manufacturing operations. As technology advances and integration becomes easier, their adoption will likely become a standard in the pursuit of smarter, more responsive production systems.

Overview of Generative AI Tools Used

For source and paper search I used ChatGPT, Version GPT-4o.

I enhanced the abstract with ChatGPT. My input was as follows: “Enhance the following abstract: More and more companies are interested in optimizing their production lines and factories. Digital twins and process mining are key components to achieve exactly that goal. Therefore it is of great importance to understand the different versions of digital twins and the methods and ways on how to build and use them.

This thesis will give the reader a theoretical evaluation of different kinds of digital twins and what their strength and benefits are, what data is needed and how to use modern simulation tools to bring a digital twin to live.

It should be suitable for a bachelor thesis.”

The output is now the abstract. Furthermore I asked ChatGPT to translate it into German which is now the Kurzfassung.

I also enhance the conclusion with ChatGPT. My input was as follows: “Enhance this conclusion: As discussed earlier, digital twins of all kinds will attract more and more attention in modern production lines. They are effectively improving throughput, output, quality, and many more factors. Using them in trainings for employees to monitoring a production line or a whole factory, the use cases are endless. Furthermore, with the right training and enough insides on how the production process works, implementing a digital twin is achievable with little effort compared to the savings possible in terms of cost saving. Finally, it is certain to say, that digital twins can improve the production processes of a manufacturing company.

It should be suitable for a bachelor thesis.” The output is now the conclusion.

List of Figures

3.1 Projects ribbon	16
3.2 Process Modeling Library ribbon	18
3.3 Material Handling Library ribbon	19
3.4 Agent ribbon	20
3.5 Space Markup ribbon	21
3.6 Presentation ribbon	22
4.1 Production layout	24
4.2 Initial setup in AnyLogic	24
4.3 Added resource pools	25
4.4 Added schedule blocks	26
4.5 Charts	27

Bibliography

- [1] McKinsey Company. Digital twins: The next frontier of factory optimization. 2024. Accessed: 2024-11-20.
- [2] Celonis. How process mining enables the digital twin of an organization (dto). 2024. Accessed: 2024-11-20.
- [3] Buro Happold. Digital twins and the importance of data. 2024. Accessed: 2024-11-20.
- [4] David Jones, Chris Snider, Aydin Nassehi, Jason Yon, and Ben Hicks. Characterising the digital twin: A systematic literature review. *CIRP Journal of Manufacturing Science and Technology*, 29:36–52, 2020.
- [5] Diego M. Botín-Sanabria, Adriana-Simona Mihaita, Rodrigo E. Peimbert-García, Mauricio A. Ramírez-Moreno, Ricardo A. Ramírez-Mendoza, and Jorge de J. Lozoya-Santos. Digital twin technology challenges and applications: A comprehensive review. *Remote Sensing*, 14(6), 2022.
- [6] Alexander Barbie and Wilhelm Hasselbring. From digital twins to digital twin prototypes: Concepts, formalization, and applications, 2024.
- [7] Roberto Argolini, Federico Bonalumi, Johannes Deichmann, and Stefania Pellegrinelli. Digital twins: The key to smart product development. *McKinsey & Company*, 2023.
- [8] Mark Crawford. 7 digital twin applications for manufacturing. *ASME*, 2021.
- [9] Triax Technologies. Digital twins in manufacturing: Top 6 applications. *Triax Technologies Blog*, 2023.
- [10] N-iX. Digital twins in manufacturing: a comprehensive overview. *N-iX Insights*, 2024.
- [11] Toobler. Top 11 digital twin examples transforming industries. *Toobler Blog*, 2024.
- [12] ASME. 7 digital twin applications for manufacturing. *ASME Topics & Resources*, 2021.

- [13] Author(s) Name(s). Digital twins for supply chains: Main functions, existing applications, and research opportunities. In *Proceedings of the IEEE Conference on [Conference Name]*, pages 1–8, 2024.
- [14] Zhiyuan Li, Xuesong Mei, Zheng Sun, Jun Xu, Jianchen Zhang, Dawei Zhang, and Jingyi Zhu. A reference framework for the digital twin smart factory based on cloud-fog-edge computing collaboration. *Journal of Intelligent Manufacturing*, 2024.
- [15] MTM ASSOCIATION e. V. The process language methods-time measurement (mtm), 2024.
- [16] AnyLogic’s website.
- [17] AnyLogic for Academia.
- [18] AnyLogic 2023 in review.
- [19] Simulation-based Railway Design Tool for Sugar Cane Logistics Optimization.